Matt Acarregui

https://www.linkedin.com/in/matt-acarregui | https://www.github.com/macarreg | macarregui22@gmail.com | (206) 900-4771

EDUCATION

Cal Poly, San Luis Obispo, CA
B.S. in Computer Science

LANGUAGES & TOOLS

Languages: Python, C, Java, JavaScript, C++
Frameworks & Libraries: React, Node.js, Flask, Pandas, NumPy, scikit-learn, NLTK, SpaCy
Databases & Cloud: MongoDB, Azure, AWS

RELEVANT COURSEWORK

- Artificial Intelligence (LLMs, Computer Vision, Neural Networks)
- Speech & Language Processing (language models in python)
- Knowledge-Based Systems
- Data Science (predictive modeling using Pandas and scikit-learn libraries)
- Operating Systems & Systems Programming
- Programming Languages (Parsers, Compilers, Grammars in Racket)
- Full-Stack App Development (React)

RESEARCH EXPERIENCE

CAHSI Research Alliance, San Luis Obispo, CA | December 2025 - June 2026

- Worked with a professor to improve LLM serving by implementing a request hierarchy for requests, specifically on heterogeneous GPU clusters.

Senior Project Research, San Luis Obispo, CA | September 2025 - March 2026

- Co-authored an academic literature review on transportation networks and studied their future applications.

PROJECTS

Knowledge-based SOMA Solver | Full-Stack

- Developed an interactive 3D model of the SOMA grid and the available pieces in JavaScript for easy user interface.
- Python backend checks for validity of puzzles, helps with hints, and connects to a Supabase DB to keep solutions.
- The program allows users to sign up and submit puzzle solutions to compete on a public leaderboard.
- Deployed via Render https://somacube.onrender.com/

Cal Poly Prop Hunt | Full-Stack

- Worked with a team to build a property listing and review website using React and NodeJS
- Database setup with MongoDB, and deployed with Microsoft Azure.

Coreference Resolution Tracker | Full-Stack

- Developed a Python-based coreference resolution system benchmarked against FastCoref.
- Links noun chunks to their respective entity using features extracted with Spacy's NLP tools.
- Simple web app which displays the clusters of references generated by each model
- Jupyter notebook which shows embedding space of clusters as well as Precision, Recall, and F1 scores for each model.

Predicting Tik-Tok "Virality" | Back End

- Used a large HuggingFace dataset (2 million rows) to develop a model to predict whether a Tik-Tok video is viral.
- Created text embeddings of video descriptions and hashtags to cluster videos by content type.
- Integrated Scikit-learn tools to determine important video features such as duration, time posted, verified users, etc.